FIITJEE

JEE(Main), 2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – I
(Main)

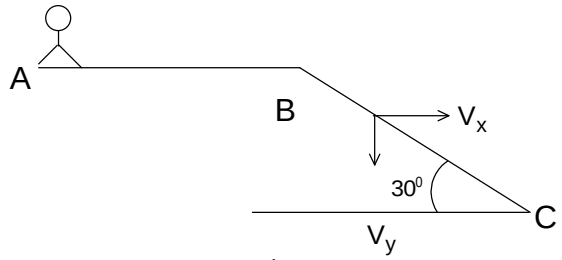
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	B	31.	B	61.	B
2.	A	32.	A	62.	A
3.	A	33.	D	63.	A
4.	C	34.	A	64.	B
5.	B	35.	D	65.	A
6.	D	36.	C	66.	B
7.	A	37.	A	67.	A
8.	B	38.	C	68.	A
9.	B	39.	B	69.	B
10.	A	40.	B	70.	A
11.	A	41.	B	71.	C
12.	A	42.	B	72.	D
13.	A	43.	C	73.	A
14.	D	44.	B	74.	B
15.	A	45.	D	75.	A
16.	B	46.	D	76.	C
17.	C	47.	D	77.	C
18.	D	48.	B	78.	C
19.	A	49.	D	79.	A
20.	C	50.	A	80.	D
21.	D	51.	A	81.	D
22.	A	52.	D	82.	C
23.	A	53.	C	83.	C
24.	B	54.	B	84.	A
25.	A	55.	C	85.	D
26.	D	56.	D	86.	C
27.	A	57.	A	87.	D
28.	D	58.	C	88.	D
29.	A	59.	A	89.	C
30.	D	60.	B	90.	B

Physics

PART – I

SECTION – A

1. $\overrightarrow{V_R} = V_x i + V_y i$



For path AB $V_x = 5\ ms^{-1}$

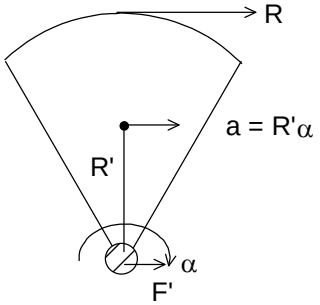
For path BC velocity of rain $\perp$r to BC should be zero hence

$V_x \sin 30^o = V_y \cos 30^o \quad V_y = \frac{5}{\sqrt{3}} ms^{-1}$

$V_R = \sqrt{V_x^2 + V_y^2} = \frac{10}{\sqrt{3}} ms^{-1}$

2. since $\overrightarrow{V_{AB}}$ and $\overrightarrow{R_{BA}}$ are antiparallel to each other initial separation is minimum at initial state

3. $\tau = I\alpha \Rightarrow \alpha = \frac{FR}{mR^2}.2 = \frac{2F}{mR}$



Position of C.O.M = $\frac{2R}{3}.\frac{\sin\left(\frac{\pi}{6}\right)}{\left(\frac{\pi}{6}\right)} = \frac{2R}{\pi} = R'$

Acceleration of C.O.M = $\frac{4F}{m\pi}$

Hence $F + F' = \frac{4R}{m\pi}.m$

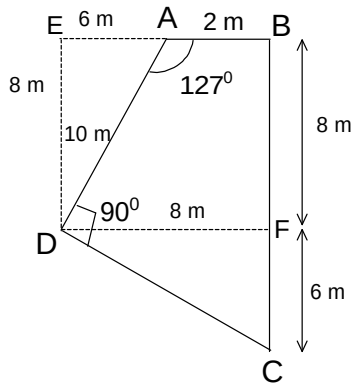
$F' = \left(\frac{4-\pi}{\pi}\right)F$

4. $a = -Kx \Rightarrow K = 4\ s^{-2}$

Then $\int v dv = K \int x dx$

$v = 8 ms^{-1}$

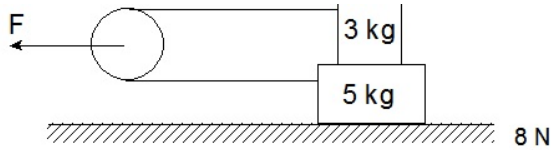
5. portion DEA and DFC are identical hence moment of inertia of ABCD is same as ABFDEA



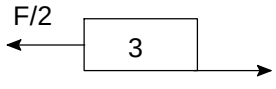
$$I = 2.\frac{M \times (BE)^2}{3}$$

$$= 512\ kgm^2$$

6. To start motion F > 8 M



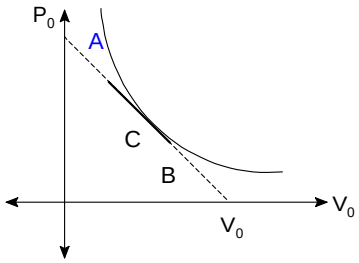
Acceleration = $\frac{F-8}{8}$



$$\frac{F}{2} - f_r = 3\left(\frac{F-8}{8}\right) \Rightarrow \frac{F}{8} + 3 = f_r$$

$$\frac{F_{max}}{8} + 3 = f_{Lm} = 6N \Rightarrow F_{max} = 24Nx$$

7. the point where adiabatic curve touch the line AB the nature of process will change from endo thermic to exothermic.

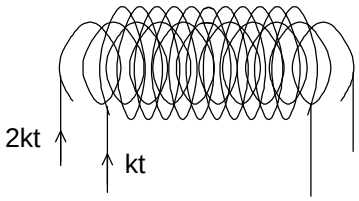


$$-\frac{P_0}{V_0} = -\gamma\frac{P}{V}$$

$$-m = -\frac{\gamma(-mv + P_0)}{V} \qquad m = +\frac{P_0}{V_0}$$

$$mv(1+\gamma) = \gamma P_0 \qquad V = \frac{\gamma P_0}{m(1+\gamma)} = \frac{5}{8}V_0$$

8. $\phi = \mu_0 nkt\pi r^2 + \mu_0 n2kt.\pi R^2$



$$E.2\pi r = \frac{d\phi}{dt} = \mu_0 nk\pi\left(2R^2 + r^2\right)$$

9. the point x = a and b are stable equilibrium point.

10. $$Fx - \mu_0 m_1 gx - \frac{1}{2}kx^2 = 0$$

$$x = \frac{\mu_0 m_2 g}{k}$$

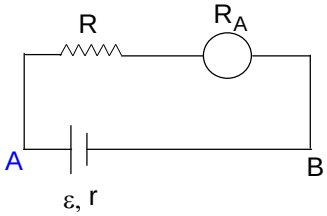
11. $$F_1 = \frac{2mv}{\Delta t} \qquad F_2 = \frac{2mv\cos 30^o}{\Delta t}$$

Hence $F_1 > F_2$

12. since mass and specific heat of two different bodies are same

$$h_f = \frac{h_1 + h_2}{2} = h_1 = h_2$$

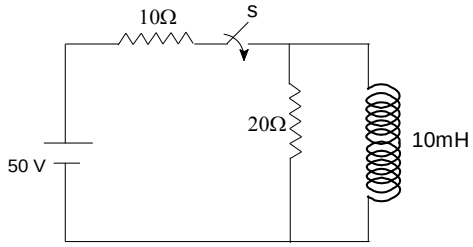
13. $$\mathrm{I} = \frac{\varepsilon}{R + R_A + r}$$



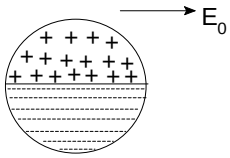
$$V = \varepsilon - \frac{\varepsilon_r}{\left(R + R_A + r\right)} = \varepsilon - \mathrm{I}r = \frac{\varepsilon\left(R + R_A\right)}{\left(R + R_A + v\right)}$$

14. at steady state 5A current through inductor hence P.D across indicator at t = 0

$20 \times 5 = 100$ V



15. Dipole moment $\left(\frac{\sigma\pi R^2}{2}\right)\left(2.\frac{4R}{3\pi}\right)$



$\frac{4R^3\sigma}{3}$

$f_r = Ma$

$$\frac{4R^3\sigma}{3}E - f_r R = \frac{MR^2}{2}\alpha$$

$$\alpha = \frac{a}{R}$$

For pure rolling

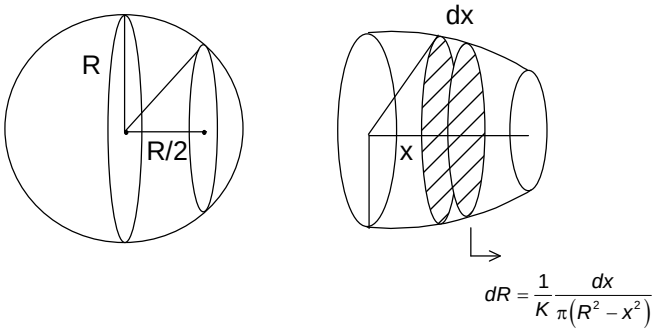
$$a = \frac{8}{9}\frac{\sigma E R^2}{M}$$

16. current across inductor and capacitor are out of phase and same magnitude hence only resistance will be operated.

$2.5\sqrt{2}\ A$

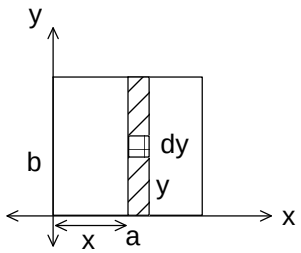
17. $$\int \frac{1}{K}.\frac{dx}{\pi\left(R^2 - x^2\right)}$$

$$R_T = \frac{1}{K\pi}\int \frac{dx}{\left(R^2 - x^2\right)} = \frac{1}{2K\pi R}\ln(3)$$



$$dR = \frac{1}{K}\frac{dx}{\pi\left(R^2 - x^2\right)}$$

18. $$dm = \left(\frac{\alpha_0 x}{ab}\right)y\,(dy)(dx)$$



$$\vec{R} = (x, y)$$

$$\frac{\int \vec{R}.dm}{\int dm} = \frac{\int \frac{\alpha_0 xy}{ab}(dxdy)\left(x\hat{i} + y\hat{j}\right)}{\int \frac{\alpha_0 xy}{ab}dx.dy}$$

$$\frac{\int \frac{\alpha_0 xy}{ab} x\, dx.dy\hat{i}}{\int \frac{\alpha_0 xy}{ab} dx,yd} + \frac{\int \frac{\alpha_0 xy}{ab} dx.dy\,(y)\,\hat{j}}{\int \frac{\alpha_0 xy}{ab} dx.dy}$$

$y = COM$ $\quad$ $x =$

$dy =$ $\quad$ $dx =$

$$\frac{\int_0^a x^2 dx}{\int_0^a x dx}\hat{i} + \frac{\int_0^b y^2 dy}{\int_0^b y dx}\hat{j}$$

$$\frac{2a}{3}\hat{i} + \frac{2b}{3}\hat{j}$$

19. $\vec{M} = 10\left\{100\times10^{-4}\hat{i} + 50\times10^{-4}\hat{j} + 50\times10^{-4}\hat{k}\right\}$

$= \left(10\hat{i} + 5\hat{j} + 5\hat{k}\right)\times10^{-3}\, Am^2$

$\vec{B} = 2\hat{i} - 3\hat{j} + \hat{k}$

$\vec{\tau} = \vec{M}\times\vec{B} = 10\times\sqrt{20}\times10^{-2}\, Nm = \sqrt{20}\times10^{-1}\, Nm$

20. C

21. Distance between central bright and first minimum,

$y = \frac{D\lambda}{d}$

$\frac{y_1}{y_2} = \frac{\lambda_1}{\lambda_2}$

22. A

23. stopping potential for a and b are same. Hence $f_a = f_b$ but current I_b is more than I_r at zero potential.

24. $I_2 = I_0\left(1 - e^{-\mu x}\right)$ $I = \frac{I_0}{2}$, $x = 1.5\ cm^{-1}$

$\mu = 0.462\ cm^2 / g$

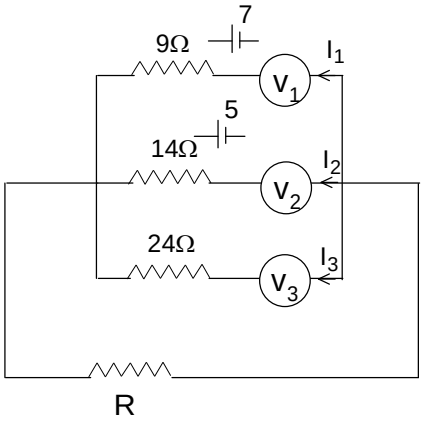
25. A

26. $R = \sigma T^4 \Rightarrow T^4 = \frac{R}{\sigma}$

$\lambda T = b$

27. A

28. since voltmeter are identical



$\frac{I_1}{I_2} = \frac{3}{2}$ *let* $I_2 = x$ $I_1 = \frac{3}{2}x$

$9 \times \frac{3}{2}x - 14x + 5 - 7.5 = 0$

$x = -5.0A = I_2$

$-7.5A = I_1$

$\frac{27x - 28x}{2} = 2.5$

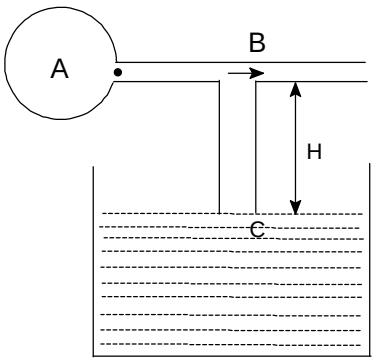
Hence resistance of voltmeter is 1 Ω

$25I_3 = -7.5 - 7.5 \times 9 = -75$

$I_3 = -3A$

Hence V_3 = 3.0 V

29. $P_B + \frac{1}{2}\sigma V^2 = P_A + P$

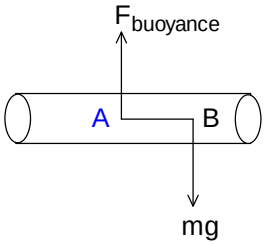


$P_B = \frac{P}{A} + P - \frac{1}{2}\sigma V^2$

$= \frac{P}{A} - \rho g H$

$V = \sqrt{\frac{2(P + \rho g H)}{\sigma}}$

30. $F_{Buoyancy} = \pi r^2 \ell \rho g$



Torque = $\frac{\pi r^2 \ell^2 \rho g}{4}$

$\alpha = \frac{\pi^2 \ell^2 \rho g r^2}{4I}$

Chemistry

PART – II

SECTION – A

31. $\frac{1}{2}\frac{d[B]}{dt} = k_1[A] \Rightarrow \frac{d[B]}{dt} = 2k_1[A]$

$\frac{d[C]}{dt} = k_2[A]$

$\Rightarrow \frac{[B]}{[C]} = \frac{2k_1}{k_2} = 1$

$\Rightarrow \%[B] = \frac{1}{1+1} \times 100 = 50\%$

32. $B_{12} = 12$ corners

Icosahedral = 20 faces.

33. Nylon – 2 – Nylon – 6

-copolymer of glycine and amino caproic acid

-biodegrable.

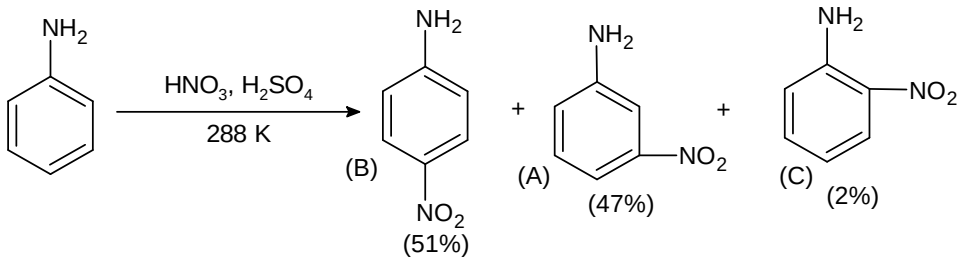
34. Average (P – P) bond enthalpy

$= \frac{1265 - 59}{6} = 201\,kJ/mol$

35. Self reduction:

$Cu_2O + Cu_2S \longrightarrow Cu + SO_2$

36.



Order of basic nature: A > B > C

37. $\text{Anode}(-) \Rightarrow \underset{1\,atm}{H_2(g)} \longrightarrow \underset{10^{-5.5}M}{2H^+(aq.)} + 2e^-$

$\text{Cathode}(+) \Rightarrow \underset{x\,M}{2H^+(aq.)} + 2e^- \longrightarrow \underset{(1\,atm)}{H_2(g)}$

$0.118\,V = 0 - \frac{0.059}{2}\log_{10}\frac{(10^{-5.5})^2}{(x)^2} \Rightarrow x = 10^{-3.5}M$

$\Rightarrow pH = 3.5$

38. Reaction involved in developer:

$K_2Fe^{II}(C_2O_4)_2 + AgBr \longrightarrow KFe^{III}(C_2O_4)_2 + Ag\downarrow + KBr$

39. $CO + 3H_2 \xrightarrow{Ni} CH_4 + H_2O$

$$CO + 2H_2 \xrightarrow{ZnO-Cr_2O_3} CH_3OH$$
$$CO + H_2 \xrightarrow{Cu} HCHO(g)$$

40. $Z < 1$ means attraction forces are dominating $\Rightarrow$ a is considerable, b can be negligible at low temperature and low pressure.

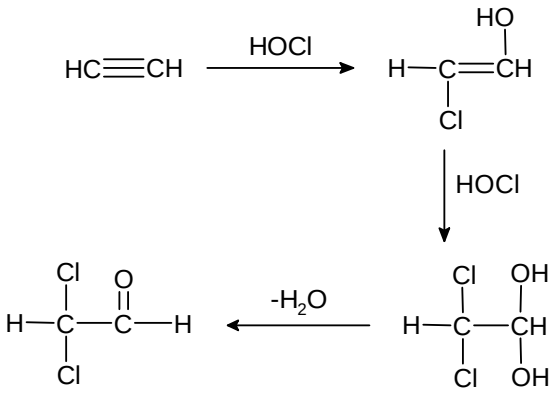
41. $X = NO_2$, $Y = N_2O$

42. Organic compound having C, N and S give blood colour in Lassaigne's test with neutral $FeCl_3$, due to formation of $Fe(SCN)_3$.

43. Chloroform and diethylether on mixing show –ve deviation from Raoult's law, in which V.P. decreases and b.p. increases.

44. E.N difference $\Rightarrow O - X > O - H$
$\Rightarrow H - O - X \longrightarrow H - O^- + X^+$
$\Rightarrow$ basic nature.

45.



46. Buffer solution

$$pH = pK_{a_2} + \log\frac{\left[CO_3^{-2}\right]}{\left[HCO_3^-\right]} = (11 - \log 2) + \log\left(\frac{40}{5}\right)$$
$$= 11 + 2\log 2 = 11.6$$

47. $\Delta_0 \ \alpha$ Strength of ligand
$\left(\text{maximum for } CN^-\right)$

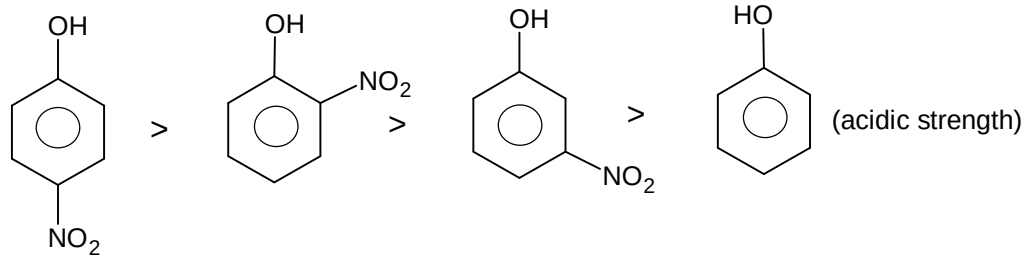
48. Representation:

$$\begin{array}{c}\text{Base}\\|\\-\text{sugar}-\text{phosphate}-\end{array}\left(\begin{array}{c}\text{Base}\\|\\\text{sugar}-\text{phosphate}\end{array}\right)_n\begin{array}{c}\text{Base}\\|\\-\text{sugar}-\end{array}$$

49. Na^+ occupies octahedral voids formed by close packing of Cl^- in ccp in which all tetrahedral voids are vacant.

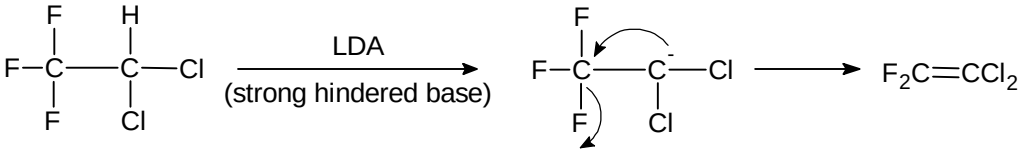
50. $ClF_3 \Rightarrow$ T – shaped

51. Acidic strength: $\propto \frac{1}{pK_a}$



52. Below pI, cationic species, goes to cathode.
Above pI, anionic species, goes to anode.

53. $Mg + C \xrightarrow{\Delta} MgC_2 \xrightarrow{\text{high T}} Mg_2C_3 \xrightarrow{H_2O} Mg(OH)_2\downarrow + C_3H_4\uparrow$

54. 

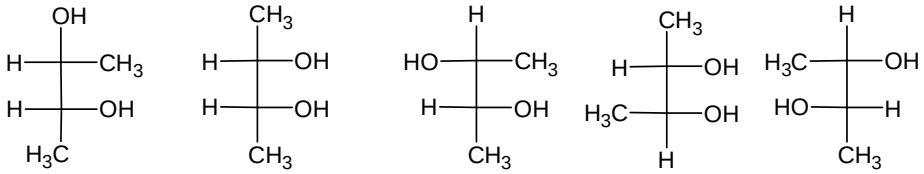
55. $T = KV \Rightarrow PV^0 = \text{constant}$

$$C = \frac{R}{\gamma - 1} - \frac{R}{n-1} = \frac{R}{\frac{7}{5} - 1} - \frac{R}{0-1} = \frac{5R}{2} + R$$

$$= \frac{7R}{2}$$

56. $Cu^{+2} + \left[Fe(CN)_6\right]^{-4} \longrightarrow \underset{\text{Chocolate brown/reddish brown ppt.}}{Cu_2\left[Fe(CN)_6\right]}$

57. For s-orbital, $\ell = 0 \Rightarrow$ no nodal plane
For d_{z^2} – orbital, $\ell = 2 \Rightarrow$ 2 conical node, no nodal plane.

58. 

59. Classical smog is a reducing mixture, while photochemical smog has high concentration of oxidizing agents.

60. Cannizaro reaction involves attack of OH^- on carbonyl carbon and H^- shift on another carbonyl carbon, no carbanion intermediate.

Mathematics

PART – III

SECTION – A

61. Let $f(x) = \sin x + 2x - kx^2 - kx$

$f'(x) = \cos x + 2 - 2kx - k$

$f''(x) = -\sin x - 2k$

The inequality is trivially true for k < 0

For $k \geq 0$, we have $f''(x) < 0$

The curve y = f(x) is concave downwards

$\Rightarrow f\left(\frac{\pi}{2}\right) \geq 0 \Rightarrow k \leq \frac{4(\pi+1)}{\pi(\pi+2)}$

62. $y = \log_{10} x$ & $z = \log_x 10 = \frac{1}{\log_{10} x} = \frac{1}{y}$

$\Rightarrow y = \frac{1}{z} \because \frac{dy}{dz} = \frac{-1}{z^2} = -\frac{1}{(\log_x 10)^2} = \frac{-(\log x)^2}{(\log 10)^2}$

63. $\lim\limits_{x\to\frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}} = \lim\limits_{x\to\frac{\pi}{4}} \frac{f(\sec^2 x)\times 2\sec^2 x.\tan x}{2x} = \frac{8}{\pi} f(2)$

64. $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} a_{n+1} = L$

$\therefore \lim\limits_{n\to\infty} a_{n+1} = \lim\limits_{n\to\infty} \sqrt{2 + a_n}$

$\Rightarrow L = \sqrt{2+L} \Rightarrow L = 2$ & $L = -1 \Rightarrow N.D$

65. Area = A = xy

$\Rightarrow A = \frac{\ln x}{x} \Rightarrow \frac{dA}{dx} = \frac{1 - \ln x}{x^2}$

$\frac{dA}{dx} = 0 \Rightarrow x = e$

$\Rightarrow$ + | - (↑ e ↓)

Hence x = e is point of maxima; $A_{max} = e.\frac{1}{e^2} = \frac{1}{e}$

66. let $I = f\left(\frac{1}{2}\right) - f\left(\frac{1}{3}\right) = \int_0^{\frac{\pi}{4}} \log_e \left(\frac{1+\frac{1}{2}\tan z}{1+\frac{1}{3}\tan z}\right) dz$

$= \int_0^{\frac{\pi}{4}} \log_e \left(\frac{3}{2}\frac{2+\tan z}{3+\tan z}\right) dz$

Replacing $z \to \frac{\pi}{4} - z$

We get $I = \int_0^{\frac{\pi}{4}} \log_e \left(\frac{3}{4} \frac{3+\tan z}{2+\tan z} \right) dz$

$\Rightarrow 2I = \int_0^{\frac{\pi}{4}} \log_e \left(\frac{9}{8} \right) dz \Rightarrow I = \frac{\pi}{8} \log_e \left(\frac{9}{8} \right)$

67. $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

$I_1 = \int_{\sin^2 t}^{1+\cos^2 t} x.f\left(x(2-x)\right)dx = \int_{\sin^2 t}^{1+\cos^2 t} (2-x) f\left(x(2-x)\right)dx = 2\int_{\sin^2 t}^{1+\cos^2 t} f\left(x(2-x)\right) - I_1$

$\Rightarrow 2I_1 = 2\int_{\sin^2 t}^{1+\cos^2 t} f\left(x(2-x)\right)dx \Rightarrow 2I_1 = 2I_2 \Rightarrow \frac{I_1}{I_2} = 1$

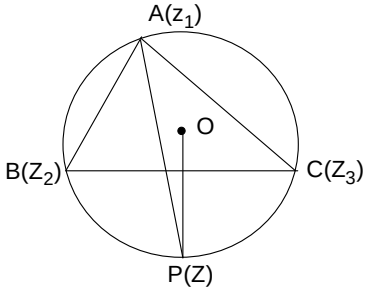
68. $x\left(y^2.e^{xy} + e^{\frac{x}{y}} \right)dy = y\left(e^{\frac{x}{y}} - y^2.e^{xy} \right)dx$

$\Rightarrow y^2 (xdy + ydx) e^{xy} = e^{\frac{x}{y}} (ydx - xdy)$

$\Rightarrow e^{xy} d(xy) = e^{x/y} \frac{ydx - xdy}{y^2} = e^{\frac{x}{y}}.d\left(\frac{x}{y}\right)$

$\Rightarrow e^{xy} = e^{\frac{x}{y}} + c$

69. since OP $\perp$ BC



$\Rightarrow \arg\left(\frac{z}{z_3 - z_2} \right) = -\frac{\pi}{2}$

70. you can have W, W, B or W, B, W or B, W, W

Required probability = $\frac{3}{4}.\frac{2}{4}.\frac{3}{4} + \frac{3}{4}.\frac{2}{4}.\frac{1}{4} + \frac{1}{4}.\frac{2}{4}.\frac{1}{4} = \frac{9}{32} + \frac{3}{32} + \frac{1}{32} = \frac{13}{32}$

71. $x^2 + \alpha x + 2 = \alpha x - 1$ (Let $\alpha x - 1 \geq 0$)

$\Rightarrow x^2 + 3 = 0$ (No solution)

Now,

$x^2 + \alpha x + 2 = 1 - \alpha x \left(\alpha x - 1 < 0\right)$

$\Rightarrow x^2 + 2\alpha x + 1 = 0$

Let $f(x) = x^2 + 2\alpha x + 1$

$\Rightarrow D \geq 0 \Rightarrow \alpha \in (-\infty,-1] \cup [1,\infty)$

If both roots –ve, then $f(0) > 0$ & $\frac{-b}{2a} \leq 0 \Rightarrow \alpha \geq 0$

So for $\alpha \geq 1$ both roots of this equation are –ve & $\alpha \leq 1$ at least one root is +ve and for –ve value afx $\alpha x - 1 < 0$ for positive values of x.

72. $|AB| \geq |PA - PB|$

So PA – PB is maximum when P is collinear with AB.

Equation of $AB \Rightarrow y - 1 = \frac{-1}{2}(x - 0)$

$\Rightarrow x + 2y = 2 \rightarrow (i)$

& solve with $4x + 3y + 9 = 0 \rightarrow$ (ii)

(I) & (ii)

$P \equiv \left(\frac{-24}{5}, \frac{17}{5}\right)$

73. A

74. As eccentric angle $\theta \in (0, 2\pi)$

$\Rightarrow |\theta_1| + |\theta_2| = \pi \Rightarrow \theta_2 = \pi - \theta_1$

$P \equiv (5\cos\theta, 4\sin\theta) Q \equiv (5\cos\theta_2, 4\sin\theta_2) or\ Q(5\cos\theta, 4\sin\theta)$

$\Rightarrow PQ \equiv y = 4\sin\theta_1$

75. equation of director circle.

76. $\left(1 - \frac{s-b}{s-a}\right)\left(1 - \frac{s-c}{s-a}\right) = 2 \Rightarrow a^2 = b^2 + c^2$

77. solving the planes taking 3 at a time vertices of tetrahedron are

$0 \equiv (0,0,0), A \equiv (1,1,-1), B \equiv (1,-1,1), C \equiv (-1,1,1)$

Volume of tetrahedron = $\frac{1}{3}$ × (area of base) (height)

78. if the points be A, B, & C respectively then

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{a} - 2\vec{b} + \lambda\vec{c} - (2\vec{a} - \vec{b} + 3\vec{c}) = -\vec{a} - \vec{b} + (\lambda - 3)\vec{c}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \mu\vec{a} - 5\vec{b} - (2\vec{a} - \vec{b} + 3\vec{c}) = (\mu - 2)\vec{a} - 4\vec{b} - 3\vec{c}$

Points are collinear if $\overrightarrow{AB} = t\overrightarrow{AC}$

$\Rightarrow \lambda = \frac{9}{4}, \mu = -2$

79. put $\ln x = t$.

80. Plane containing the line of intersection of the given planes is

$(x - y - z - 4) + \lambda(x + y + 2z - 4) = 0$

$s.e. (\lambda + 1)x + (\lambda - 1)y + (2\lambda - 1)z - 4(\lambda + 1) = 0$

Vector normal to it

$V = (\lambda + 1)\hat{i} + (\lambda - 1)\hat{J} + (2\lambda - 1)\hat{K}$

Vector along the line of intersection of the planes
$2x+3y+z-1=0$ & $x+3y+2z-2=0$ is

Given by $\vec{n}=3\left(\hat{i}-\hat{J}+\hat{K}\right)$

& $\vec{n}$ is parallel to the pane $\rightarrow$ (i)

$\Rightarrow \vec{n}.\vec{v}=0 \Rightarrow \lambda=\frac{-1}{2}$

& hence the required plane is
$x-3y-4z-4=0$

$\Rightarrow |A+B+C|=11$

81. $\sin^{-1}$ is defined for [-1, 1] $\Rightarrow a=0$

$\therefore$ x + y = $\sin^{-1}1+\cos^{-1}1-\tan^{-1}1=\frac{\pi}{4}$

Clearly image about x – axis will be x – y = $\pi/4$.

82. $\cos 4x+6-7cos2x=0$

$\Rightarrow (cos2x-1)(2cos2x-5)=0$

$\Rightarrow cos2x=\frac{5}{2}$ & $\cos 2x=1$

2x = 2nπ $\Rightarrow$ x = nπ
Sum in [0,314].
$=\pi[1+2+.....+100]-100\pi=5050\pi-100\pi=4950\pi$

83. we have g(x) = f(x) Sinx $\rightarrow$ (i)
Diff. w.r to x.
g'(x) = f(x) Cosx + f'(x) Sinx $\rightarrow$ (ii)
Again diff.
g"(x) = f(x) (-Sinx)+ f'(x) Cosx + f'(x) Cosx +f"(x) Sinx
$\Rightarrow g''(-\pi)=2f'(-\pi).\text{Cos}(-\pi)=2\times 1\times -1=-2$
Hence $g''(-\pi)=-2$

84. we have $\frac{3x+6y}{k}=1\rightarrow$ (i)
& $2x^2+2xy+3y^2-1=0\rightarrow$ (ii)
Now Homogenising (2) with the help of (1),
We get

$$2x^2+2xy+3y^2-\left(\frac{3x+6y}{k}\right)^2=0$$

$$\Rightarrow K^2\left(2x^2+3x+3y^2\right)-\left(3x+6y\right)^2=0$$

Now coff. of x^2 + coff. of y^2 = 0
$\Rightarrow K^2=9 \Rightarrow K=\pm 3$

85. D

86. Let the amount received by sons be Rs. X, Rs. Y & Rs. Z respectively, then
$x\le y+z=101-x$
i.e $2x\le 101\Rightarrow x\le 50, y\le 50$ & $z\le 50$ & $x+y+z=101$
$\therefore$ Coff. of x^{101} in the expansion of $(1+x+x^2+.....+x^{50})^3$
$={}^{1\times 103}C_{101}-3.{}^{52}C_{50}={}^{103}C_2-3.{}^{52}C_2$

87. $z_1, z_2, z_3 \ldots\ldots z_k = \cos\left(\sum_{n=1}^{n=k} \frac{1}{n(n+1)(n+2)}\right) + i\sin\left(\sum_{n=1}^{n=k} \frac{1}{n(n+1)(n+2)}\right)$

$= \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$

88. Since the tangent at any point $\forall$ x (a, b) lies below the curve
$\therefore$ the curve is concave upward in the interval (a, b)
$\therefore$ f"(x) > 0 $\forall$ x $\in$ (a, b)

89. $x^2y^3 = (x+y)^n$

$2xy^3 + 3x^2y^2\frac{dy}{dx} = n(x+y)^{n-1}\left(1+\frac{dy}{dx}\right)$

$\Rightarrow 5xy^3 = \frac{n(x+y)^n}{x}\left\{\because \frac{dy}{dx} = \frac{y}{x}\right\}$

$\Rightarrow 5(x+y)^n = n(x+y)^n$

$\Rightarrow n = 5$

90. $\alpha + \beta = -a \rightarrow$ (i)

$\alpha.\beta = a+1 \rightarrow$ (ii)

Adding (I) & (ii)

$\alpha + \beta + \alpha\beta = 1 \Rightarrow (\alpha+1)(\beta+1) = 2$

α & β are integral

$\alpha + 1 = 1, \beta + 1 = 2$

$\alpha + 1 = -1, \beta + 1 = -2$

$\alpha = 0, \beta = 1$

Or

$\alpha = -2 \,\&\, \beta = -3$

$\Downarrow$

$a = -1 \text{ or } 5$